3.7 Solving Nonlinear Systems of Equations

Name _____ Period ____ Date _____

Algebra II 3.7 · Student sheet

I can…

- solve a system with a linear and a quadratic equation by graphing.
- solve nonlinear systems by substitution and elimination.
- solve f(x) = g(x) by graphing or algebra.

Vocabulary · fill in the missing word

System of nonlinear equations — _____

Line and parabola — can _____ in 0, 1, or 2 points

Two parabolas — can meet in 0, 1, 2, or _____ many points

f(x) = g(x) — the _____ where the graphs of y = f(x) and y = g(x) meet

Part 1 · How many crossings?

Graph $y = x^2$ and $y = x + k$ in Desmos with a slider for k.

k	−2	−¼	0	2
Number of intersection points				

For what value of k does the line just touch the parabola? _____

Part 2 · Notes: fill in the blanks

A line and a parabola can meet in _____ points.

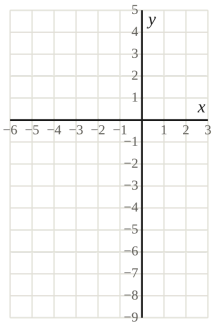
Substitution: solve one equation for _____, then _____.

Always find _____ coordinates of each solution.

Part 3 · Examples (work them with the class)

Example 1

Solve the system by graphing. $y = x^2 + 4x - 4$ and $y = 2x - 5$



Example 2

Solve the system. $y = x^2 - 1$ and $y = x + 1$

Example 3

Solve the system.

$x^2 - 4x - y = -5$

$x^2 - 2x + y = 5$

Example 4

Solve $f(x) = g(x)$ for $f(x) = x^2 + 1$ and $g(x) = 3x - 1$.

Part 4 · Partner practice

Partner 1

Solve the system. $y = x^2$ and $y = -x + 6$

Partner 2

Solve $f(x) = g(x)$ for $f(x) = x^2 - 2x$ and $g(x) = x - 2$.